PLUMBING PIPING SCHEDULE:

SYSTEM	LOCATION	SIZE	PIPE MATERIAL	JOINTS	FITTINGS	HORIZONTAL HANGER SPACING (FT.)	VERTICAL SPACING (FT.)	NOTES
DOMESTIC COLD AND HOT WATER	ABOVE GRADE	1-1/4" AND SMALLER	TYPE L HARD COPPER	SOLDER / PROPRESS	WROUGHT COPPER	6	10	-
		1-1/2" AND LARGER				10	10	-
SANITARY SEWER	ALL	1-1/4" AND SMALLER	HUBLESS CAST IRON	HUBLESS	HUBLESS CAST IRON	5	15	1, 2
		1-1/2" AND LARGER	DWV HARD COPPER	SOLDER	CAST BRONZE OR WROUGHT COPPER	10	10	-
		1-1/4"				6	10	-
		ALL	PVC SHED 40 DWV	SOLVENT WELD	DWV SOCKET	4	10	3, 4
NATURAL GAS	ABOVE GRADE	1/2"	SCHD 40 BLACK IRON	THREADED, WELDED, MEGAPRESS	MALLEABLE IRON, MEGAPRESS	6	15	-
		3/4" THRU 1-1/2"				8	15	-
		2" AND LARGER		WELDED, MEGAPRESS	BUTT WELDED, MEGAPRESS	10	15	-

GENERAL NOTES:

A. ALL PRODUCTS USED IN DOMESTIC COLD AND HOT WATER PIPING SYSTEMS MUST BE NSF/ANSI 61 CERTIFIED AND BE LEAD FREE. LEAD FREE PRODUCTS MUST BE LISTED AND LABELED BY A THIRD PARTY AS LEAD FREE SUCH THAT THE WETTED SURFACE MATERIALS HAVE NO MORE THAN 0.25% LEAD CONTENT.

B. INSULATION TO BE FIBERGLASS, POLYMER FOAM, OR ELASTOMERIC PREFORMED PIPE INSULATION WITH SELF SEALING ADHESIVE LAP.

C. ANY PIPING TO BE RECEIVING FIBERGLASS INSULATION MUST ALSO HAVE VAPOR RETARDED JACKET, SEAL JOINTS WITH 2" WIDE PRESSURE SENSITIVE VAPOR RETARDED TAPE.

D. INSULATION MUST BE INSTALLED ACCORDING TO MANUFACTURER'S REQUIREMENTS.

E. PIPE HANGING AND SUPPORT COMPONENTS AND METHODS TO BE IN ACCORDANCE WITH ANSI/MSS SP-58-2009.

F. FOR INSULATED PIPING PROVIDE PIPE HANGER INSULATION SHIELD ACCORDING TO HANGER MANUFACTURER.

G. HANGER SPACING DOES NOT APPLY TO UNDERGROUND AND BELOW SLAB PIPING.

H. DUCTILE IRON AND CAST IRON PIPING MUST HAVE POLYETHYLENE ENCASEMENT WHEN INSTALLED IN CORROSIVE SOILS, VERIFY IN FIELD.

I. HUBLESS CAST IRON JOINTS MUST BE MADE WITH ASTM C1540 HEAVY DUTY SHIELDED COUPLINGS WITH CENTER STOPS.

J. PVC PIPE MUST BE SOLID WALL, FOAM CORE IS NOT PERMITTED.

K. DO NOT USE PVC PIPING IN RETURN AIR PLENUM.

NOTES:

1. AT LEAST 1 ADDITIONAL HANGER IS REQUIRED AT EACH JOINT.
2. 10' HANGER SPACING IS PERMITTED ONLY IF 10' LENGTHS OF PIPE ARE USED.
3. PIPE MATERIAL MAY BE USED WHERE PERMITTED BY AUTHORITY HAVING JURISDICTION.
4. FOR SIZES 2 INCHES AND SMALLER, A GUIDE SHALL BE INSTALLED MIDWAY BETWEEN REQUIRED VERTICAL SUPPORTS. SUCH GUIDES SHALL PREVENT PIPE MOVEMENT IN A DIRECTION PERPENDICULAR TO THE AXIS OF THE PIPE.

MINIMUM PIPE INSULATION THICKNESS

FLUID OPERATING TEMPERATURE RANGE (°F)	INSULATION CONDUCTIVITY		NOMINAL PIPE OR TUBE SIZE (IN)				
	CONDUCTIVITY BTU*IN./(H*FT2**F)	MEAN RATING TEMPERATURE, °F	< 1	1 TO < 1-1/2	1-1/2 TO < 4	4 TO < 8	≥ 8
> 350	0.32 - 0.34	250	4.5	5.0	5.0	5.0	5.0
251-350	0.29 - 0.32	200	3.0	4.0	4.5	4.5	4.5
201-250	0.27 - 0.30	150	2.5	2.5	2.5	3.0	3.0
141-200	0.25 - 0.29	125	1.5	1.5	2.0	2.0	2.0
105-140	0.21 - 0.28	100	1.0	1.0	1.5	1.5	1.5
61-104	-	82	0.5	0.5	1.0	1.0	1.0
40-60	0.21 - 0.27	75	0.5	0.5	1.0	1.0	1.0
< 40	0.20 - 0.26	50	0.5	1.0	1.0	1.0	1.0

NOTES:

1. ALL PIPING TO THE INLET OF A WATER HEATER AND PIPING CONVEYING WATER HEATED BY A WATER HEATER SHALL BE INSULATED. DOMESTIC COLD WATER SHALL BE INSULATED IN ACCORDANCE WITH THE TABLE ABOVE BASED ON FLUID TEMPERATURE DEPENDING ON GROUNDWATER AT PROJECT LOCATION.
2. FOR PIPING SMALLER THAN 1-1/2 INCHES AND LOCATED IN PARTITIONS WITHIN CONDITIONED SPACES, REDUCTION OF THESE THICKNESSES BY 1 INCH SHALL BE PERMITTED (BEFORE THICKNESS ADJUSTMENT REQUIRED IN FOOTNOTE B) BUT NOT TO A THICKNESS LESS THAN 1 INCH.
3. MISCELLANEOUS FLUIDS SUCH AS WASTE PIPING FROM ELECTRIC WATER COOLERS TO DRAINAGE, REFRIGERANT SUCTION LINES OR HOT GAS BYPASS LINES, AND COOLING COIL CONDENSATE PIPING ARE TO BE INSULATED IN ACCORDANCE WITH THE TABLE ABOVE BASED ON FLUID TEMPERATURE.

WATER HEATER SCHEDULE

MARK	PRODUCT		TANK CAPACITY	ELECTRICAL			WATER			
	MANUFACTURER	MODEL		KW	FLA	VOLT / PH / HZ	IN/OUT SIZE	DRAIN SIZE	DELTA T	TEMP.
EWH	LOCHINVAR	LDJ-10-JE	10 GAL	2.5 KW	-	208/1/60	3/4"	3/4"	78°F	140°F

NOTES: 62°F COLD WATER TEMPERATURE USED FOR CALCULATIONS

1. PROVIDE A 2 GALLON BLADDER EXPANSION TANK (WATTS PLT-5-M1 OR EQUAL) ON COLD WATER INLET, INSTALL DOWNSTREAM OF CHECK VALVE.
2. PROVIDE AUTOMATIC LEAK DETECTOR SHUTOFF SYSTEM (FLOODSTOP FS3/4NPT V4 OR EQUAL) ON DOMESTIC COLD WATER SUPPLY TO WATER HEATER. PROVIDE SENSOR LOCATED IN WATER HEATER DRAIN PAN.
   A. PROVIDE LEAK DETECTION CONTROL PANEL WITH PLUG-IN POWER CONNECTION. PROVIDE 120V RECEPTACLE AT WATER HEATER LOCATION FOR CONNECTION. REFER TO ELECTRICAL PLANS FOR RECEPTACLE AND CIRCUIT REQUIREMENTS.
   B. SYSTEM SHALL AUTOMATICALLY SHUT OFF WATER UPON LEAK DETECTION AND REQUIRE MANUAL RESET.
3. PROVIDE HEAT TRAPS ON HOT AND COLD WATER CONNECTIONS.
4. ROUTE WATER HEATER DRAIN PAN TO NEAREST FLOOR DRAIN OR MOP SINK WITH AIR GAP.

CIRCULATION PUMP SCHEDULE

MARK	SERVICE	GPM	HEAD (FT.)	MOTOR	V/PH/HZ	AMPS	RPM	BASIS OF DESIGN (OR EQUAL)	NOTES
CP	DOMESTIC HOT WATER	-	-	1 / 40	115/1/60	0.52	3250	TACO 006-ST4	ALL

PUMP OPERATION NOTES:

1. PROVIDE RECIRCULATION PUMP WITH INTEGRAL TIMER; PROVIDE SEPERATE TIMER IF NOT INCLUDED WITH PUMP. TIMER SHALL PREVENT PUMP OPERATION FOR A MINIMUM OF 20 MINUTES (ADJ.) AFTER SHUTDOWN.
2. INSTALL AQUASTAT ON HOT WATER RETURN LINE AND CONTROL PUMP OPERATION AS FOLLOWS:
   A. ACTIVATE PUMP WHEN RETURN WATER TEMPERATURE DROPS BELOW 100°F (ADJ.)
   B. DEACTIVATE PUMP WHEN RETURN WATER TEMPERATURE EXCEEDS 120°F (ADJ.)
3. PROVIDE BALANCING VALVE IN RECIRCULATION LINE AND SET FLOW TO 0.5 GPM.
4. PROVIDE CHECK VALVE IN RECIRCULATION LINE DOWNSTREAM OF PUMP OR UTILIZE PUMP WITH INTEGRAL CHECK VALVE.
5. PROVIDE 115V, SINGLE-PHASE POWER CONNECTION AT PUMP LOCATION. REFER TO ELECTRICAL PLANS FOR CIRCUITING AND CONNECTION REQUIREMENTS.

PIPE MATERIALS

CONDITION/LOCATION	MATERIAL TYPE
ABOVE GROUND WATER	PEX-A[1]
BELOW GROUND WATER	PEX-A[2]
WASTE (UNDER 2")	SCHEDULE 40 PVC PLASTIC
WASTE (ABOVE 2")	SCHEDULE 40 PVC PLASTIC
WASTE	SCHEDULE 40 PCV PLASTIC
VENT	SCHEDULE 40 PCV PLASTIC
VENT (UNDER 2")	SCHEDULE 40 PVC PLASTIC
VENT (ABOVE 2")	CAST IRON
CONDENSATE PIPING	SCHEDULE 40 PVC PLASTIC

NOTE: IF THE ABOVE MATERIALS ARE NOT PERMITTED BY LOCAL CODES, USE THE FOLLOWING PIPING MATERIALS:
- TYPE "L" COPPER FOR ABOVE GROUND WATER LINES
- TYPE "K" COPPER FOR BELOW GROUND WATER LINES
- TYPE "M" COPPER FOR WASTE AND VENT LINES UNDER 2"
- CAST IRON FOR WASTE LINES 2" AND LARGER
- SCHEDULE 40 WROUGHT IRON FOR CONDENSATE PIPING

1. USE TYPE "L" COPPER WHERE PIPING IS EXPOSED TO UV LIGHT.

2. USE TYPE "K" COPPER WHERE PIPING IS EXPOSED TO UV LIGHT.

SLOPE OF HORIZONTAL DRAINAGE PIPE

SIZE(INCHES)	MINIMUM SLOPE (INCHES PER FOOT)
2-1/2 OR LESS	1/4
3 TO 6	1/8
8 OR LARGER	1/16

NOTE: TABLE INFORMATION FROM IPC TABLE 704.1

MAXIMUM DISTANCE OF FIXTURE TRAP FROM VENT

SIZE PR TRAP (INCHES)	SLOPE (INCHES PER FOOT)	DISTANCE FROM TRAP (FEET)
1-1/4	1/4	5
1-1/2	1/4	6
2	1/4	8
3	1/8	12
4	1/8	16

NOTE: TABLE INFORMATION FROM IPC TABLE 909.1

MINIMUM SIZE OF FIXURE WATER SUPPLY PIPES

FIXTURE (1)	MIN. PIPE SIZE (INCH)
DISHWASHER, DOMESTIC (2)	1/2
DRINKING FOUNTAIN	3/8
HOSE BIBB	1/2
KITCHEN SINK (2)	1/2
LAVATORY	3/8
SINKS, SERVICE	1/2
URINAL, FLUSH TANK	1/2
URINAL, FLUSHOMETER VALVE	3/4
WALL HYDRANT	1/2
WATER CLOSET, FLUSH TANK	3/8
WATER CLOSET. FLUSHOMETER TANK	3/8
WATER CLOSET. FLUSHOMETER VALVE	1
WATER CLOSET. FONE PIECE (2)	1/2

NOTES:

1. TABLE INFORMATION GATHERED FROM IPC TABLE 604.5.

2. WHERE THE DEVELOPED LENGTH OF THE DISTRIBUTION LINE IS 50 FEET OR LESS, AND THE AVAILABLE PRESSURE AT THE METER IS 35 PSI OR GREATER, THE MINIMUM SIZE OF AN INDIVIDUAL DISTRIBUTION LINE SUPPLIED FROM A MANIFOLD AND INSTALLED AS PART OF A PARALLEL WATER DISTRIBUTION SYSTEM SHALL BE ONE NOMINAL TUBE SIZE SMALLER THAN THE SIZES INDICATED.

PIPING VOLUME AND MAXIMUM PIPING LENGTHS

PIPING SIZE (INCHES)	VOLUME (OZ/FT)	MAX. PIPING LENGTH (FEET)	
		PUBLIC LAVATORY	OTHER FIXTURES
1/4	0.33	6	50
5/16	0.5	4	50
3/8	0.75	3	50
1/2	1.5	2	43
5/8	2	1	32
3/4	3	0.5	21

NOTE: TABLE INFORMATION FROM IECC TABLE C404.5.1

PLUMBING FIXTURE SCHEDULE

MARK	DESCRIPTION	BASIS OF DESIGN	TRIM/ACCESSORIES	CONNECTION SIZES				NOTES
				CW	HW	WASTE	VENT	
SK-T	TOP-MOUNTED LOUNGE SINK	BELAY LRAD-2219	SINK DRAIN OFFSET TO BACK, PROVIDE WITH LK-35 STRAINER, MCGUIRE #1912 P-TRAP AND # 165 STOPS WITH SUPPLIES.	1/2"	1/2"	2"	1-1/2"	1, 2, 3, 4
	LOUNGE FAUCET	SYMMONS DIA S-3510-PD	1.0 GPM FAUCET, PROVIDE WITH KOHLER K-18036 0.5GPM FEMALE AERATOR.	3/8"	3/8"	-	-	3
SK-U	UNDER-MOUNT LIBRARY SINK	ELKAY DCFU1618	SINGLE BOWL CENTERED DRAIN OFFSET TO BACK, PROVIDE WITH LK-35 STRAINER, MCGUIRE #1912 P-TRAP AND #165 STOPS WITH SUPPLIES.	1/2"	1/2"	2"	1-1/2"	1, 2, 3, 4
	LIBRARY FAUCET	SYMMONS DIA S-3510-PD	1.0 GPM FAUCET, PROVIDE WITH KOHLER K-18036 0.5GPM FEMALE AERATOR.	3/8"	3/8"	-	-	3
WC	WATER CLOSET - ADA COMPLIANT	KOHLER K-3519-T-0	FLOOR MOUNTED (1.0 GPF) VITREOUS CHINA, ELONGATED BOWL, AND TANK COVER LOCKS. PROVIDE BEMIS 1655SSCT SEAT WITH SELF-SUSTAINING STAINLESS STEEL CHECK HINGE. PROVIDE KOHLER K-7637 3/8" ANGLE SUPPLY WITH STOP AND ESCUTCHEON.	1"	-	4"	2"	1, 3
LAV	LAVATORY -ADA COMPLIANT	KOHLER K-2005 COMPLETE	WALL HUNG ADA COMPLIANT LAVATORY, PROVIDE WITH K7715 OUTLET WITH TAILPIECE, ZURN Z1231 FIXTURE SUPPORT, MCGUIRE ST02LK SUPPLIES WITH STOPS AND MCGUIRE #8902 P-TRAP. INSULATE P-TRAP, STOPS AND SUPPLIES WITH "PRO-WRAP" BY MCGUIRE. MOUNT WITH RIM MAXIMUM 34" AFF.	1/2"	1/2"	2"	1-1/2"	1, 2, 3, 4
	LAVATORY - (BATTERY) FAUCET	ZURN AQUASENSE Z6915-XL	BATTERY SENSOR ACTIVATED FAUCET WITH 0.5 GPM SPOUT AERATOR. PROVIDE WITH KOHLER K-7129-A DRAIN AND SYMMONS 2-225-CK-MS MIXING BELOW SINK, ADJUSTED TO PROVIDE 105F TO HOT WATER SIDE OF FAUCET.	3/8"	3/8"	-	-	2, 3
EDF	WATER COOLER - HILOW ADA COMPLIANT	ELKAY EZSTL8C	BI-LEVEL, STAINLESS STEEL CABINET, WITH WATERWAYS MANUFACTURED OF 100% LEAD FREE MATERIAL, ZURN Z1225 FIXTURE SUPPORT, BALL VALVE STOP WITH SUPPLY, SAFETY-GUARD BUBBLER. MCGUIRE #8872 P-TRAP. FULLY INSULATE P-TRAP. MOUNT WITH LOWER SPOUT OUTLET 36" ABOVE FINISH FLOOR. PROVIDE COLOR CHART FOR ARCHITECT COLOR SELECTION. PROVIDE WITH ELKAY MODEL #LKAPREZL CANE APRON AS REQUIRED.	1/2"	-	2"	1-1/2"	1, 2, 3
MS	MOP SINK BASIN	FIAT MSK-2424	24"X24"X10" FIBERGLASS OR SIMILAR AND E77-AA BUMPER GUARD. T-35 HOSE WITH WALL HOOK, STAINLESS STEEL BACKSPLASH AND KOHLER K-8907 WITH VACUUM BREAKER. REFER TO ARCHITECTURAL ELEVAITONS FOR FRP HEIGHTS.	-	-	3"	1-1/2"	3
	WALL FAUCET	KOHLER K-837T60-4-CP	TRITON BOWE SERVICE SINK FAUCET, WALL-TO-SPOUT OUTLET WITH WALL BRACE AND PAIL HOOK.	1/2"	1/2"	-	-	3
FL	FLOOR DRAIN	J.R. SMITH #2010	6" ROUND NICKEL BRONZE GRATE. PROVIDE WITH TRAP PRIMER CONNECTION. PROVIDE WITH J.R. SMITH TRAP INSERT.	-	-	3"	-	N/A
DB	DRAIN BOX	SIOUX CHIEF MODEL #696-3F	PROVIDE DRAIN BOX WITH #696-LC LOUVERED COVER, #696-CF SECONDARY DRAINAGE FUNNEL, AND J.R. SMITH TRAP SEAL INSERT. BOX TO COME COMPLETE WITH WALL FLANGE AND LOUVER. PIPE INDIRECT IN WALL TO DRAIN BOX. COORDINATE EXACT LOCATION WITH ARCHITECT AND GC.	-	-	2"	-	N/A
WH	WALL HYDRANT	J.R. SMITH #5509-QT	PROVIDE WITH INTEGRAL BACKFLOW PREVENTER, LATCHING COVER, FREEZE-PROOF AND OF PROPER LENGTH FOR WALL IN WHICH INSTALLED, ALL BRONZE BOX. VALVE SEAT MUST BE ON BUILDING SIDE OF EXTERIOR WALL INSULATION. INSTALL WITH CENTER LINE 24" ABOVE FINISH GRADE. PROVIDE OWNER WITH ONE (1) LOOSE KEY FOR EACH WALL HYDRANT.	3/4"	-	-	-	N/A
RH	ROOF HYDRANT	JR SMITH #5906-H	PROVIDE WITH DRAIN TUBE PIPED TO NEAREST DRAIN. DRAIN PIPE TO BE COPPER TO THE NEAREST DRAIN OR JANITORS RECEPTOR.	3/4"	-	3/8"	-	N/A
RD	ROOF DRAIN	J.R. SMITH #1011	COMPLETE WITH SUMP RECEIVER AND UNDER DECK CLAMP, AND CAST IRON OR ALUMINUM DOME. PROVIDE 4" HIGH, 1/16: THICK PERFORATED STAINLESS STEEL GRAVEL TRAP AROUND DOME.	-	-	SEE PLAN	-	N/A
OD	OVERFLOW DRAIN	J.R. SMITH #1080	COMPLETE WITH SUMP RECEIVER, UNDER DECK CLAMP, 2" EXTERNAL DAM, AND CAST IRON OR ALUMINUM DOME.	-	-	SEE PLAN	-	N/A
CD	CANOPY DRAIN	ZURN ZRB185	6" DIAMETER CORNICE DRAIN DURA-COATED CAST IRON, SIDE OUTLET BODY, PLAIN BRONZE DOME AND MEMBRANE FLASHING CLAMP.	-	-	SEE PLAN	-	N/A
DSN	DOWNSPOUT NOZZLE	J.R. SMITH #1770	DOWNSPOUT NOZZLE. MOUNT 1'-0" ABV FINISH GRADE. COORDINATE EXACT LOCATION WITH ARCHITECT. ENSURE PROTRUSION IS UNDER 4" FOR ADA CLEARANCE.	-	-	SEE PLAN	-	N/A
VB	WALL VALVE BOX	OATEY 38689	STEEL WALL MOUNTED BOX WITH 1/4 TURN BRASS BALL VALVE FOR CONNECTION TO COUTERTOP DRINKING EQUIPMENT.	1/2"	-	-	-	N/A

NOTES:

1. REFER TO ARCHITECTURAL ELEVATIONS FOR MOUNTING HEIGHT AND ARCHITECTURAL DRAWINGS FOR EXACT MOUNTING POSITION OF FIXTURES.
2. FAUCET HOLES TO MATCH FAUCET SPECIFIED.
3. FIXTURE ASSEMBLY MUST BE APPROVED BY AND INSTALLED PER ADA REQUIREMENTS.
4. PROVIDE INSULATION KIT ON ALL ADA FIXTURES EXPOSED SUPPLIES AND DRAIN TRAP WITH TRUBRO LAV-GUARD IN WHITE.

* BASIS OF DESIGN PRODUCTS ARE INDICATED FOR REFERENCE ONLY. COMPARABLE PRODUCTS WILL BE ACCEPTED PROVIDED COMPLIANCE WITH TECHNICAL SPECIFICATION REQUIREMENTS.

CLEANOUT SCHEDULE

MARK	DESCRIPTION	BASIS OF DESIGN	TRIM/ACCESSORIES	DRAIN
FCO	CLEAN-OUT (FLOOR)	J.R. SMITH #4023S(INSIDE) #4240 (OUTSIDE)	INTERIOR 3/4" DIA, MEDIUM-DUTY CAST IRON FLOOR MOUNTED CLEANOUT WITH ROUND ADJUSTABLE SCORIATED SECURED NICKEL BRONZE TOP. TAPER THREAD. PROVIDE PVC PLUG WHERE PERMITTED BY LOCAL CODES; OTHERWISE, PROVIDE BRONZE PLUG.	PER PLAN
WCO	CLEAN-OUT (WALL)	J.R. SMITH #4422C	WALL MOUNTED CLEANOUT WITH ACCESS COVER, TAPER THREAD. PROVIDE PVC PLUG WHERE PERMITTED BY LOCAL CODES; OTHERWISE, PROVIDE BRONZE PLUG.	PER PLAN

* BASIS OF DESIGN PRODUCTS ARE INDICATED FOR REFERENCE ONLY. COMPARABLE PRODUCTS WILL BE ACCEPTED PROVIDED COMPLIANCE WITH TECHNICAL SPECIFICATION REQUIREMENTS.

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	07/02/2026	PERMIT SET

Client:
FIFTH THIRD BANK

Project Number: 26032
Project Name:
FIFTH THIRD BANK WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:
PLUMBING TABLES AND SCHEDULES

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P-101

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